

An
INTERNSHIP REPORT
On

University ChatBot

Submitted

By

Gopeshkumar Rabadiya
(Enrollment Number:211430116017)

Guided By:

Prof. Priya Chaturvedi

An

Internship Report

Submitted to



Gujarat Technological University

In fulfillment for the award of degree of Bachelor of Engineering

in

IT

ACADEMIC YEAR – 2025



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

Pakwan, Behind Rajpath Club Gate to Sindhu Bhavan Road,
Sarkhej - Gandhinagar Highway, AEC Char Rasta, Ahmedabad, Gujarat 380054



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

CERTIFICATE

This is to certify that the Internship report submitted along with the project entitled Internship in **HYPX ECOSYSTEM LLP** has been Completed by **Gopeshkumar Rabadiya** under my guidance in complete fulfilment for the Bachelor of Engineering in **IT** Branch, 8th Semester of **Gujarat Technological University**, Ahmedabad during the academic year 2025.

Date:

Place: NEW LJiet, Ahmedabad.

Signature and Name of Guide

Prof. Priya Chaturvedi
Assistant Professor ,
IT Department,
NEW LJiet (143), Ahmedabad.

Signature and Name of H.O.D.

Dr. Gayatri Pandi.
IT Department,
NEW LJiet (143), Ahmedabad.

Signature and Name of Principal

Dr. Gayatri Pandi.
NEW LJiet (143), Ahmedabad

Seal of Institute



GUJARAT TECHNOLOGICAL UNIVERSITY

CERTIFICATE FOR COMPLETION OF ALL ACTIVITIES AT ONLINE PROJECT PORTAL
B.E. SEMESTER VIII, ACADEMIC YEAR 2024-2025

Date of certificate generation : 17 April 2025 (10:36:45)

This is to certify that, ***Rabadiya Gopeshkumar Harsukh*** (Enrolment Number - 211430116017) working on project entitled with ***University ChatBot*** from ***INFORMATION TECHNOLOGY*** department of ***New L.J Institute of Engineering and Technology*** had submitted following details at online project portal.

Internship Project Report	Completed
---------------------------	-----------

Name of Student : Rabadiya Gopeshkumar
Harsukh

Name of Guide : M r . D I V Y A N G
PRAHLADBHAI MAKWANA
Mr. PRIYA SURENDRA
CHATURVEDI

Signature of Student : _____

*Signature of Guide : _____

Disclaimer :

This is a computer generated copy and does not indicate that your data has been evaluated. This is the receipt that GTU has received a copy of the data that you have uploaded and submitted as your project work.

*Guide has to sign the certificate, Only if all above activities has been Completed.



20th January 2025

Date: 15/04/2025

To Whom It May Concern,

This is to certify that Mr. Rabadiya Gopeshkumar Harsukh, residing in Ahmedabad, Gujarat, has successfully completed his internship at HypX Ecosystem LLP as a Software Developer Intern. His internship tenure was from 20th January 2025 to 19th April 2025, during which he worked from our Ahmedabad office.

During the internship period, Gopesh contributed to various real-world projects, primarily focusing on event ticketing systems and user experience optimization. He was actively involved in the development process and worked with the following technologies:

- FASTAPI (Python)
- Next.js and React
- TypeScript
- PostgreSQL
- Prisma ORM
- AWS Cloud Services

He demonstrated a good understanding of our technology stack, maintained professional conduct, adhered to coding standards, and collaborated effectively with the development team. His contribution during the internship was valuable and appreciated by the team.

We wish him all the best for his future endeavors and career in software development.

Best Regards,
Hitarth Thummar
Director
HypX Ecosystem LLP





NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

DECLARATION

We hereby declare that the Internship report submitted along with the Internship entitled Project **University ChatBot** submitted in Complete for Bachelor of Engineering in **IT** Branch to **Gujarat Technological University**, Ahmedabad, is a bonafide record of original Internship work Completed by me at **HYPX ECOSYSTEM LLP** under the supervision of **External Guide Hitarth Thummar** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of Student

Gopeshkumar Rabadiya

Signature of Student

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my **External guide Hitarth Thummar** for continuously guiding me at the company and answering all my doubts with patience. I would also like to thank **Dr. Gayatri Pandi (IT Department)** for motivating me every time whenever I get confused, I would also like to thank my **Internal Guide Prof. Priya Chaturvedi** for helping me through my internship by giving me the necessary suggestions and advice along with their valuable coordination in completing this Internship.

I also thank my parents, friends and all the members of the family for their precious support and encouragement which they had provided in completion of my work. In addition to that, I would also like to mention the company personnel who gave me the permission to use and experience the valuable resources required for the Internship.

Thus, in conclusion to the above said, I once again thank the staff members of **HYPX ECOSYSTEM LLP** for their valuable support in completion of the Internship.

Thank You

Name of Student

Gopeshkumar Rabadiya

Enrollment Number

211430116017

Date:

Signature of Student

TABLE OF CONTENTS

CHAPTERS	PAGE NO.
Title Page	i
Certificate Page	ii
Internship Certificate	iv
Declaration	v
Acknowledgements	vi
Table of Contents	vii
Abstract	x
CHAPTER 1	01
INTRODUCTION OF PROJECT & COMPANY PROFILE	
1.1 Introduction	01
1.1.1 Company Profile	02
1.1.2 Company Products	02
1.1.3 Company Mission and Vision	02
1.2 Introduction of the Project	03
1.2.1 Purpose of the Project	03
1.2.2 Scope of the Project	03
1.2.3 Motivation	04
CHAPTER 2 SYSTEM REQUIREMENTS	05
2.1 Introduction	06
2.2 Hardware Requirements	06
2.3 Software Requirements	06
2.4 External Dependencies	07
CHAPTER 3 WORKSHEET REPORT	08
3.1 WORK SHEET REPORT (15 DAYS)	09
CHAPTER 4 FRONT END OF SYSTEM	16
4.1 Introduction	17

4.2 Objectives of the Front End	17
4.3 Tools & Technologies Used	17
4.4 Key UI Features	17
4.5 User Flow	18
CHAPTER 5 BACK END OF SYSTEM	19
5.1 Introduction	20
5.2 Objectives of the back end	20
5.3 Technologies Used	20
5.4 Functional Modules	21
5.5 Backend Workflow	21
5.6 Security and Error Handling	22
CHAPTER 6 SYSTEM DESIGN	23
6.1 System Design	24
6.2 Architectural Overview	24
6.3 System Components	24
6.4 Data Flow Diagram (DFD)	25
6.5 Use case Diagram	26
6.6 Sequence Diagram	26
6.7 System Flowchart	27
6.8 Conclusion	27
CHAPTER 7 DATA DICTIONARY	28
7.1 Introduction	29
7.2 Database Overview	29
7.3 Tables and Collections	29
7.3.1 MongoDB Collection	29
7.3.2 SQLite Tables	30
7.4 Indexing and performance Optimization	31
7.5 Conclusion	31

CHAPTER 8 TESTING	32
8.1 Introduction	33
8.2 Type of testing performed	33
8.3 Test Cases	34
8.4 Tools Used for Testing	35
8.5 Bug Fixes and Improvements	35
8.6 Conclusion	35
CHAPTER 9 SNAPSHOT OF WEBSITE	36
9.1 How To Run a Server	37
9.2 Main Page of Our Site	37
9.3 Login/Sign Up Page	38
9.4 Admin Panel	38
9.5 Chat Analysis	39
9.6 Example	39
9.7 Settings	40
CHAPTER 10 ADVANTAGES	41
10.1 Advantages	42
10.2 Limitations	43
CHAPTER 11 CONCLUSION AND FUTURE ENHANCEMENT	44
11.1 Conclusion	45
11.2 Future Enhancement	46
BIBLIOGRAPHY	48
a. Course Outcomes	48
b. Books	49
c. Web Reference	50

University ChatBot

Enrollment No.: 211430116017

Student Name: Gopeshkumar Rabadiya

NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

(College Code:143)

Semester: VIII, IT Department

ABSTRACT

The project titled "**University Chatbot**" is an AI-powered assistant developed to streamline communication and support within an academic institution. Built using **Streamlit**, **Python**, **MongoDB**, **SQLite**, and integrated with **Google Gemini Pro**, this chatbot provides students with quick access to academic information, college services, and FAQs in a conversational manner.

The primary aim of the chatbot is to enhance user experience by reducing the need for manual intervention from college staff and providing immediate responses to common queries. Leveraging Natural Language Processing (NLP) and Gemini Pro's advanced capabilities, the bot understands user intent and responds accurately.

This project was carried out during the final semester internship at **HYPX ECOSYSTEM LLP**, where the focus was on integrating cutting-edge AI models with web-based deployment for real-world utility in the education domain.

This report elaborates on the design, development, and deployment of the system, highlighting the challenges faced, the technology stack used, and the performance outcomes. The chatbot acts as a digital bridge between students and college administration, promoting efficiency, accessibility, and innovation in academic support systems.

CHAPTER: 1

INTRODUCTION OF PROJECT & COMPANY PROFILE

1.1. Introduction

1.1.1. Company Profile

1.1.2. Company Products

1.1.3. Company Mission and Vision

1.2. Introduction of the Project

1.2.1. Purpose of the Project

1.2.2. Function Requirements

1.2.3. Problems in Existing System

1.2.4. Main Modules

1.1. Introduction

1.1.1. Company Profile

HypX is a technology-driven company specializing in artificial intelligence, machine learning, and digital transformation solutions. The company aims to bridge the gap between businesses and cutting-edge AI solutions by offering innovative products and services tailored to various industries. With a strong emphasis on research and development, HypX is committed to **advancing AI technology** and delivering impactful solutions that enhance productivity and user experience.

Website: <https://www.hypx.in/>

1.1.2. Company Products

HypX provides a range of AI-driven solutions, including:

- **AI Chatbots** – Intelligent conversational agents for businesses and educational institutions.
- **Data Analytics Services** – Advanced data processing and predictive analytics.
- **Machine Learning Solutions** – Custom models tailored to industry-specific needs.
- **Automation Tools** – AI-powered automation solutions to enhance efficiency.

1.1.3 Company Mission & Vision

- **Mission:**

To **revolutionize industries** by leveraging AI and machine learning technologies, making cutting-edge solutions accessible and impactful for businesses and individuals.

- **Vision:**

To become a global leader in AI-driven innovation by continuously developing solutions that drive digital transformation and enhance user experiences.

1.2 Introduction of the Project

1.2.1. Purpose of the Project

The purpose of this project is to:

- Create a centralized digital assistant that is accessible 24/7.
- Reduce the workload of administrative staff by automating repetitive queries.
- Provide a smart, efficient, and scalable support system using AI.
- Enhance student experience through instant and accurate responses.

1.2.2. Scope Of The Project

This chatbot is specifically tailored for educational institutions, with a primary focus on universities and colleges. The scope includes:

- Integration of Large Language Models (LLMs) to understand complex queries.
- A web-based user interface using **Streamlit** for ease of use.

- Backend support using **MongoDB** and **SQLite** for session management and data storage.
- Future scope for integration with Learning Management Systems (LMS), email services, and voice interfaces.

1.2.3. Motivation

The inspiration for developing this chatbot stems from the need to digitize and automate student support systems. During the internship at **HYPX ECOSYSTEM LLP**, the idea of building an intelligent, accessible, and responsive assistant for university use cases was conceptualized. With the growing capabilities of AI models like Gemini Pro, it became feasible to create a functional assistant capable of understanding and resolving user queries accurately.

CHAPTER: 2

SYSTEM REQUIREMENTS

2.1 Introduction

2.2 Hardware Requirements

2.3 Software Requirements

2.4 External Dependencies

2.1 Introduction

- Before developing any software application, it is essential to define its system requirements. These include both hardware and software components necessary to ensure smooth installation, execution, and maintenance. The University Chatbot was designed with a focus on accessibility, low resource consumption, and seamless integration with AI models and databases.

2.2 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB or more
Storage	At least 500 MB free space
Display	13” display, 1024x768 resolution
Internet Connection	Required for Gemini API access

2.3 Software Requirements

Software	Description
Operating System	Windows / macOS / Linux (Ubuntu preferred)
Programming Language	Python 3.10+
Frontend Framework	Streamlit
AI Integration	Google Gemini Pro API
Database	MongoDB Atlas (for cloud storage), SQLite (local)
IDE / Code Editor	VS Code / PyCharm / Jupyter Notebook

Software	Description
Web Browser	Chrome / Firefox / Edge
Python Libraries	streamlit, requests, sqlite3, pymongo, etc.

2.4 External Dependencies

- **Google Gemini Pro** – For handling natural language queries and generating intelligent responses.
- **MongoDB Atlas** – Used for storing user sessions and historical queries securely in the cloud.
- **SQLite** – Used for lightweight, local database operations.
- **Streamlit** – Used for creating the web-based interface.

CHAPTER: 3

WORK SHEET REPORT

3.1 Work Sheet Report (15 DAYS)

3.1 Work Sheet Report (15 DAYS):

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
1	20/01/2025 To 24/01/2025	Introduction to the project, understanding the chatbot workflow and UI, and setting up the development environment.	Successfully set up the development environment with Streamlit and Python. Understood Gemini integration scope.	Project base structure and Streamlit installation completed.
2	27/01/2025 To 31/01/2025	Setup basic Streamlit app and explore Gemini Pro integration.	Implemented basic Streamlit app structure, established Gemini Pro API input/output connection.	Gemini Pro tested in basic scenario; API key management done.

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
3	03/02/2025 To 07/02/2025	Develop chatbot front-end interface with user input and response section.	Created minimal chatbot UI and integrated Gemini response flow.	
4	10/02/2025 To 14/02/2025	Add session management, prepare for database storage using SQLite.	User chat sessions created, local storage logic with SQLite implemented.	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
5	17/02/2025 To 21/02/2025	Start working with MongoDB for cloud database integration.	Configured MongoDB Atlas, created MongoDB connection module, and tested document insertion.	
6	24/02/2025 To 28/02/2025	Finalize backend response logic and Gemini API calls.	Implemented final response formatting, error handling, and message context flow.	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
7	03/03/2025 To 07/03/2025	Polish the UI and improve UX using Streamlit components.	Integrated cards, markdown, text styling, and layout enhancements.	
8	10/03/2025 To 14/03/2025	Testing chatbot interaction and error flows.	Added logging, performed unit tests for API and database failures.	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
9	17/03/2025 To 21/03/2025	Add authentication (login/signup) and session validation.	Built simple login/signup logic using Streamlit's session state.	
10	24/03/2025 To 28/03/2025	Implement additional features like chatbot reset, history view, etc.	Added reset chat button, view previous messages, and download conversation.	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
11	31/03/2025 To 04/04/2025	Prepare for project documentation and code cleanup.	Removed test files, added docstrings, organized repository for GitHub.	
12	07/04/2025 To 11/04/2025	Final presentation preparation and deployment testing.	Final version tested, screenshots taken, documentation prepared.	

<u>SUGGESTED 15 DAYS WORK SHEET REPORT</u>				
Student Name:		Gopeshkumar Rabadiya		
Enrollment No:		211430116017		
Internship/Project Title		University ChatBot		
Tools and Technologies		• Front End: Streamlit • Back End: Python, Streamlit, Gemini Pro API • Database: MongoDB (Cloud), SQLite (Local) • Development Tools: Visual Studio Code		
Company/ Organization Name		HYPX ECOSYSTEM LLP		
Student's Activity Details:				
Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
13	14/04/2025 To 18/04/2025	Make a detailed Report and PPT	Reviewed report and presentation	

CHAPTER: 4

FRONT END OF SYSTEM

4.1 Introduction

4.2 Objectives of the front end

4.3 Tools & Technologies Used

4.4 Key UI Features

4.5 User Flow

4.1 Introduction

- The front end of the University Chatbot system is developed using **Streamlit**, a Python-based open-source framework that enables rapid web app development. The interface is designed to provide a user-friendly and interactive experience for students, allowing them to chat with the AI assistant in real-time.

4.2 Objectives of the Frontend

- To create a simple and responsive UI for user interaction.
- To enable users to input queries and receive responses from the Gemini Pro model.
- To manage user sessions and display chat history in a readable format.
- To enhance the usability of the system with visually appealing layouts.

4.3 Tools & Technologies Used

- **Streamlit**: For building the entire UI layout, including forms, text inputs, and chat display.
- **Python**: Core language for integrating frontend with backend APIs and session control.
- **Streamlit Components**: Used for containers, markdown rendering, sidebar, buttons, etc.

4.4 Key UI Features

- **Text Input Area**: Allows users to type their queries to the chatbot.

- **Chat Display Panel:** Displays both user messages and bot responses in a chat-like format.
- **Sidebar:** Shows user session options such as Reset Chat, Logout, etc.
- **Buttons:** For submitting queries, refreshing the page, or downloading conversation logs.
- **Dynamic Layout:** Responsive design using columns and containers to adjust layout across devices.

4.5 User Flow

1. User logs in or signs up.
2. User types a question into the input field.
3. Frontend sends the query to the backend (Gemini API).
4. Response is received and displayed immediately.
5. Chat is stored in session and optionally saved to database.

CHAPTER: 5

BACK END OF SYSTEM

5.1 Introduction

5.2 Objectives of the back end

5.3 Technologies Used

5.4 Functional Modules

5.5 Backend Workflow

5.6 Security and error handling

5.1 Introduction

- The back end of the University Chatbot system serves as the logic and data processing layer. It handles communication between the user interface and the AI model (Google Gemini Pro), manages user sessions, stores data in databases (MongoDB and SQLite), and processes chatbot responses in real time.

5.2 Objectives of the Back End

- To process and forward user queries to the Gemini Pro model.
- To manage user sessions and handle authentication.
- To store chat interactions securely in databases.
- To return AI-generated responses to the front end efficiently.
- To maintain code modularity and ensure robust error handling.

5.3 Technologies Used

Component	Technology
Programming Language	Python
AI/LLM Integration	Google Gemini Pro API
Database (Cloud)	MongoDB (via PyMongo)
Database (Local)	SQLite (via sqlite3)
Backend Logic Handling	Python modules/functions

Component	Technology
API Handling	requests library

5.4 Functional Modules

- **Gemini Integration Module**

Sends user input to the Gemini API and retrieves the AI-generated response. Uses token-based authentication for secure access.

- **Database Handler**

MongoDB: Stores user details and chat history for persistent storage.

SQLite: Stores session data and user activity during each local session.

- **Session Manager**

Manages login state, tracks current user, and ensures messages are stored per session.

- **Response Formatter**

Formats the Gemini response into a readable message format before sending it to the UI.

5.5 Back end Workflow

1. User submits input via frontend.
2. Input is passed to backend using Streamlit's event-handling.
3. Query is sent to the Gemini Pro API using an authenticated POST request.
4. Response is received and formatted.
5. Chat is logged into MongoDB and SQLite with timestamps.
6. Response is sent back to frontend for display.

5.6 Security and error handling

- API keys are securely stored and accessed through environment variables.
- Input validation is applied to prevent empty or invalid messages.
- Try-except blocks handle API failures or network errors gracefully.
- User data is securely stored with proper document structure in MongoDB.

CHAPTER 6

SYSTEM ANALYSIS AND SYSTEM DESIGN

6.1 System Design

6.2 Architectural Overview

6.3 System Components

6.4 Data Flow Diagram (DFD)

6.5 Use case Diagram

6.6 Sequence Diagram

6.7 System Flowchart

6.8 Conclusion

6.1 System Design

System design defines the architecture, components, and interactions within the University Chatbot system. This chapter presents the structural and functional aspects, including data flow and system architecture.

6.2 Architectural Overview

The system follows a Client-Server Architecture, where:

- The Frontend (Client) built with Streamlit interacts with the user.
- The Backend (Server) processes requests, interacts with the Gemini Pro API, and manages MongoDB & SQLite databases.

6.3 System Components

1. User Interface (UI) Layer

- Built with **Streamlit** for interaction.
- Sends queries and displays chatbot responses.

2. Processing Layer (Backend)

- Handles API requests, processes responses.
- Manages authentication and session storage.

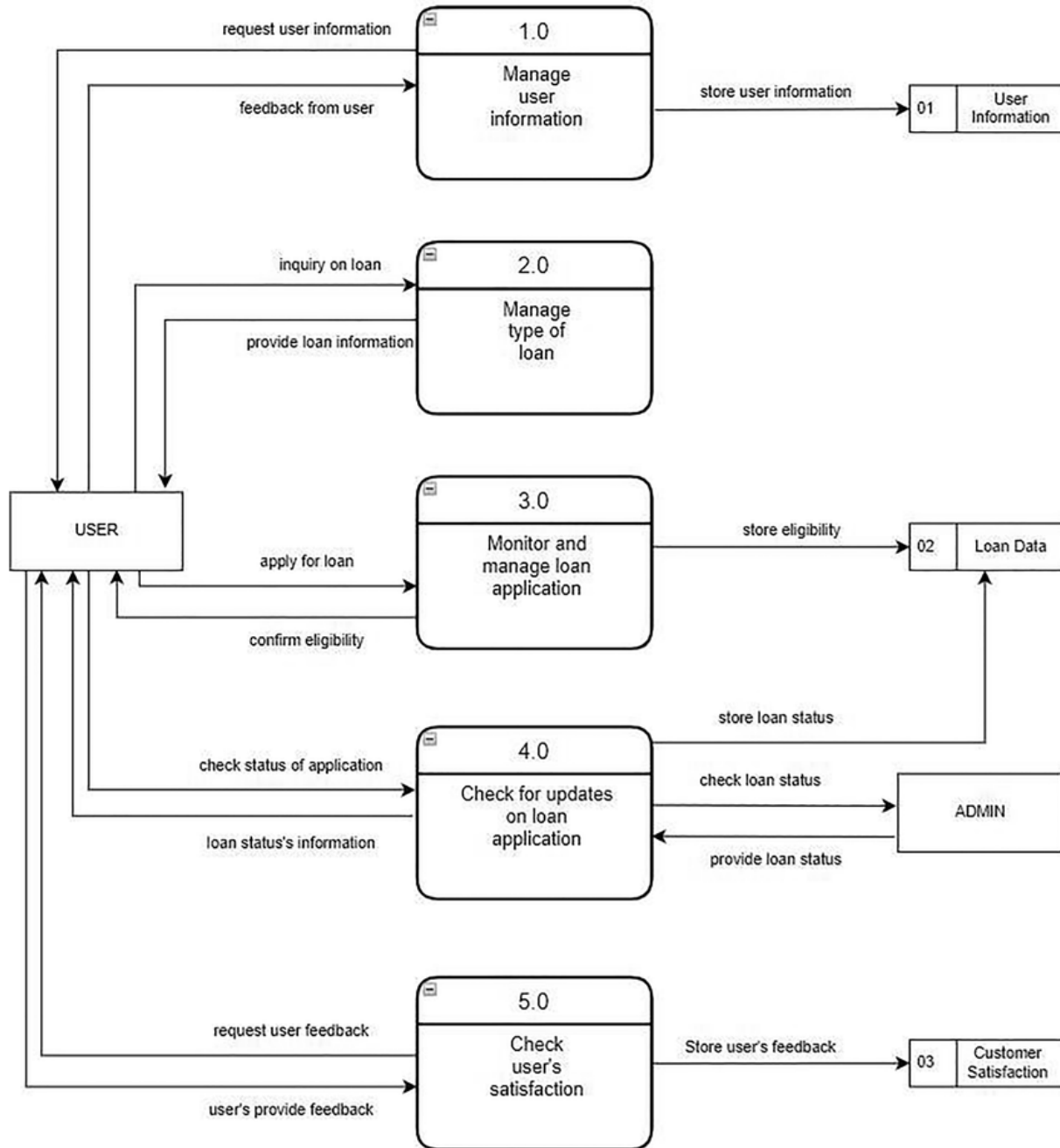
3. Database Layer

- **MongoDB (Cloud Storage)**: Stores chat history, user data.
- **SQLite (Local Storage)**: Handles temporary session-based storage.

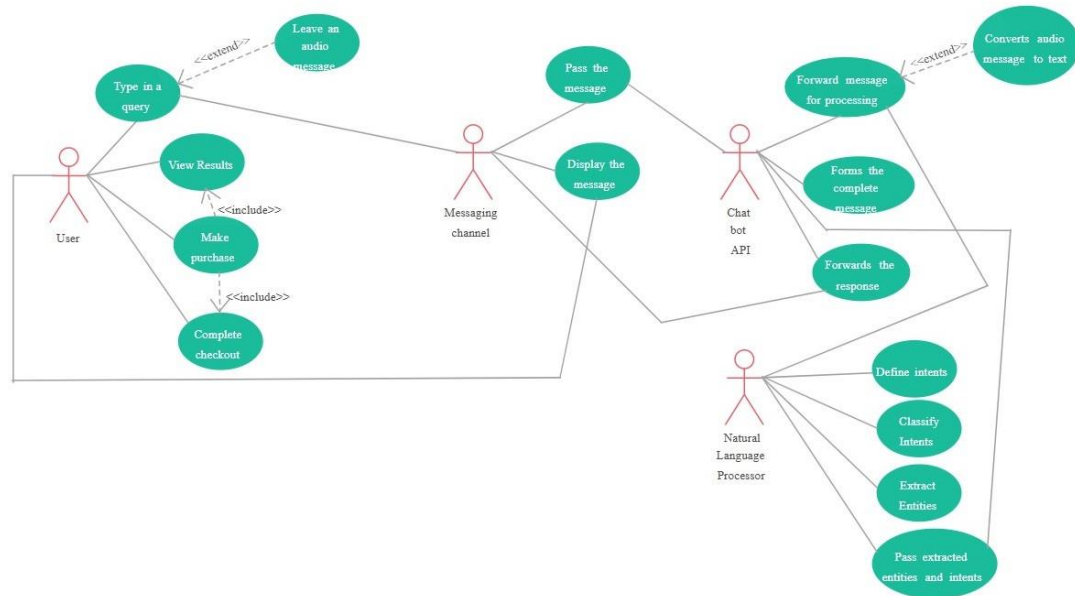
4. AI Model Layer

- **Google Gemini Pro API** processes NLP queries and generates responses.

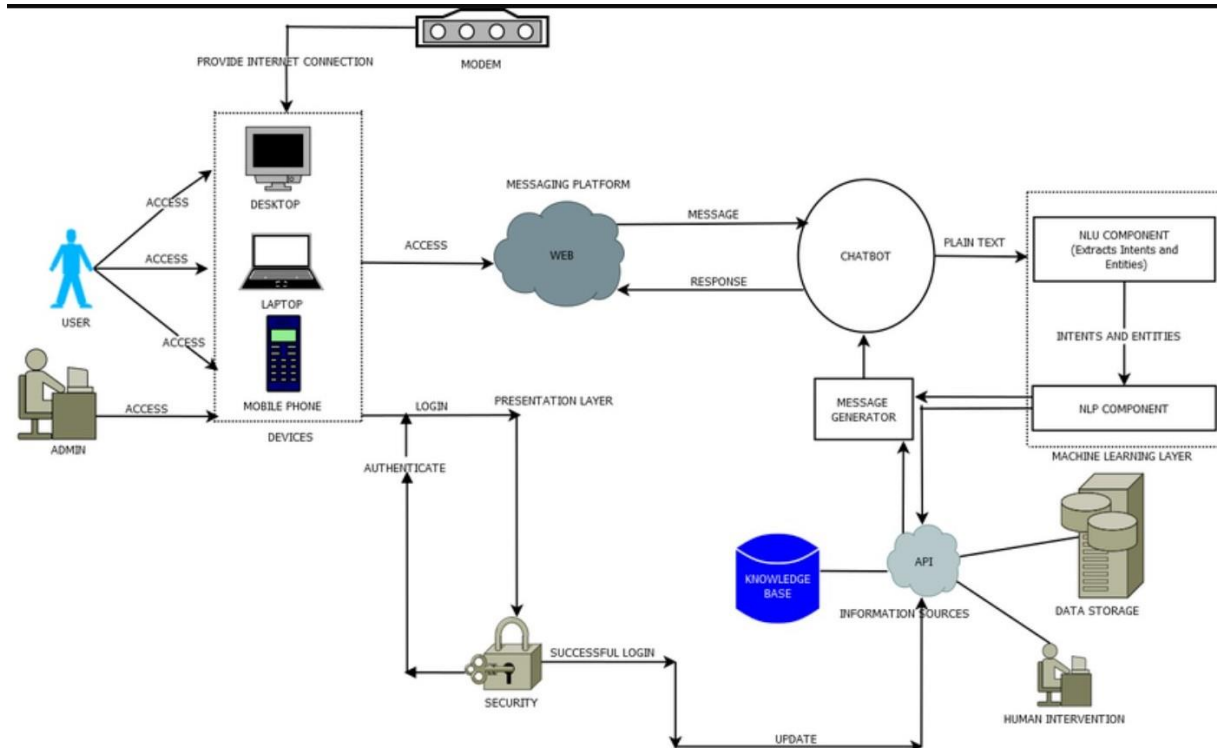
6.4 Data Flow Diagram (DFD)



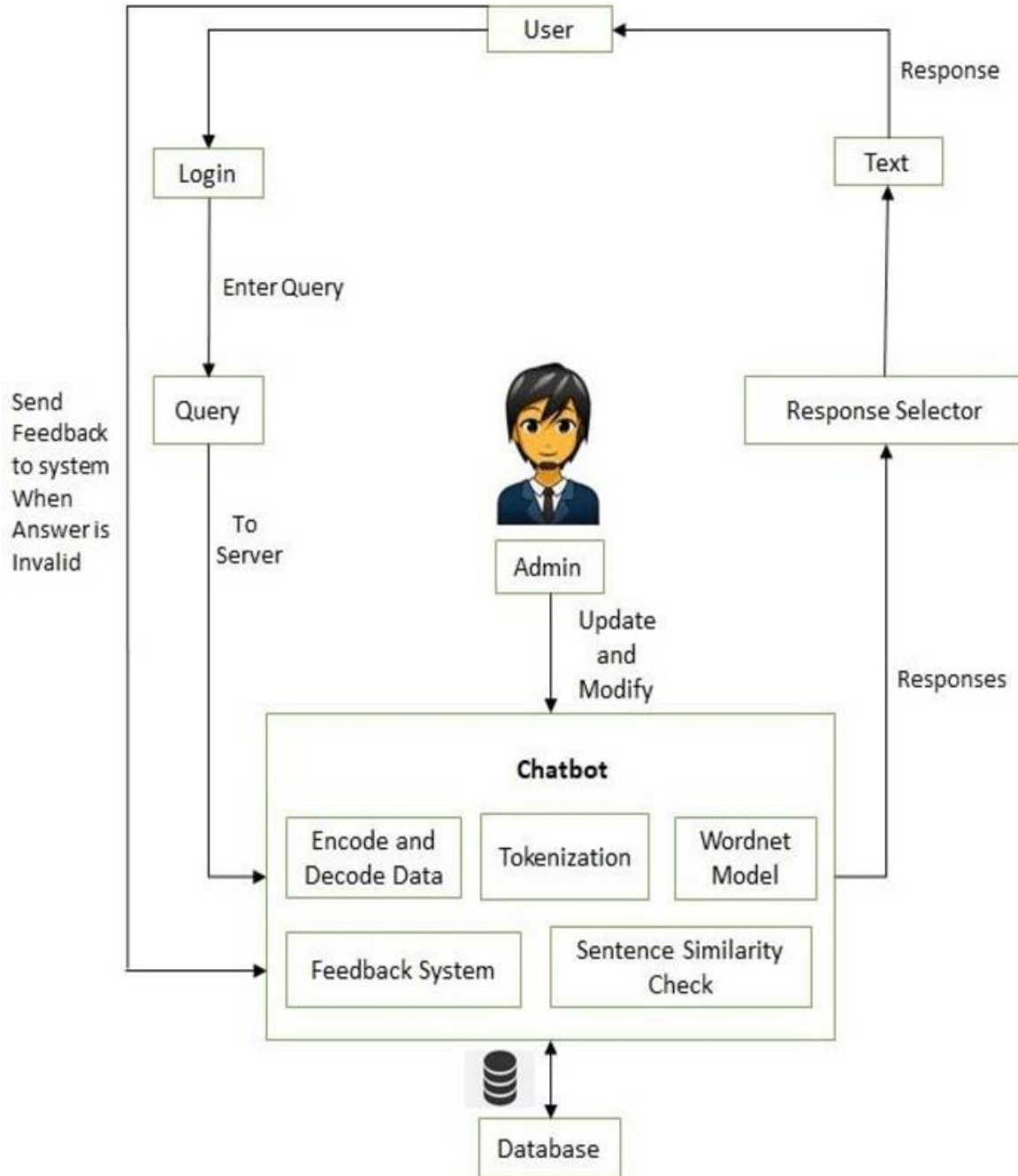
6.5 Use case Diagram



6.6 Sequence Diagram



6.7 System Flowchart



6.8 Conclusion

The system design ensures an efficient, scalable, and interactive chatbot that integrates AI capabilities with database management. The architecture is modular, allowing easy upgrades and future enhancements.

CHAPTER: 7

DATA DICTIONARY

7.1 Introduction

7.2 Database Overview

7.3 Tables and Collections

7.3.1 MongoDB Collection

7.3.2 SQLite Tables

7.4 Indexing and performance Optimization

7.5 Conclusion

7.1 Introduction

- The **Data Dictionary** defines the structure of the database used in the University Chatbot system. It provides details about the tables, fields, data types, and constraints applied to ensure data integrity and efficient retrieval.

7.2 Database Overview

The system uses two databases:

- **MongoDB (Cloud Database):** Used for storing user data, chat history, and system logs.
- **SQLite (Local Database):** Used for session-based storage and quick access during user interactions.

7.3 Tables And Collections

7.3.1 MongoDB Collections

MongoDB is a NoSQL database, so it stores data in **collections** instead of tables.

Collection Name	Field Name	Data Type	Description
users	_id	ObjectId	Unique identifier for the user
	name	String	Full name of the user
	email	String	Email address of the user

Collection Name	Field Name	Data Type	Description
	password_hash	String	Encrypted password for authentication
	created_at	Timestamp	Date and time of account creation
chat_history	_id	ObjectId	Unique ID for each chat session
	user_id	ObjectId	Reference to the user in the users collection
	message	String	User's query or chatbot response
	role	String	Indicates if the message is from the user or bot
	timestamp	Timestamp	Time when the message was sent

7.3.2 SQLite Tables

SQLite is used for local storage of temporary session data.

Table Name	Field Name	Data Type	Description
sessions	session_id	INTEGER (PK)	Unique session ID
	user_id	INTEGER (FK)	References users table in MongoDB
	start_time	DATETIME	Timestamp when session started
messages	message_id	INTEGER (PK)	Unique message ID
	session_id	INTEGER (FK)	References sessions table
	message_text	TEXT	Stores the chatbot/user messages
	sender	TEXT	Indicates whether message is from user or chatbot
	created_at	DATETIME	Timestamp of message

7.4 Indexing And Performance optimization

- MongoDB indexes on _id, user_id to optimize query performance.
- SQLite foreign keys ensure relational integrity.

7.5 Conclusion

- This **data dictionary** ensures **clarity, consistency, and efficiency** in database management for the University Chatbot system. Proper indexing

and structured storage help improve system **scalability** and **query performance**.

CHAPTER:8

TESTING

8.1 Introduction

8.2 Type of testing performed

8.3 Test Cases

8.4 Tools Used for Testing

8.5 Bug Fixes and Improvements

8.6 Conclusion

8.1 Introduction

- System testing ensures that all components of the University Chatbot are functioning correctly, both individually and as an integrated system. Testing verifies reliability, performance, usability, and correctness of the system.

8.2 Types of testing performed

Type	Description
Unit Testing	Verifies individual Python functions/modules (e.g., Gemini response handler, DB connection).
Integration Testing	Ensures modules (frontend, backend, databases, API) work together smoothly.
Functional Testing	Checks that the chatbot gives correct answers, stores chat logs, and handles sessions.
UI Testing	Ensures UI responsiveness and proper display in Streamlit.
Database Testing	Verifies insertion, retrieval, and deletion of data from MongoDB and SQLite.
API Testing	Confirms Google Gemini Pro API returns valid responses under various inputs.

User Acceptance Testing (UAT)	Real users tested the system to validate usability and expected behavior.
--------------------------------------	---

8.3 Test Cases

Sample Test Case 1: User Authentication

Test Case	Login with valid credentials
Expected Result	User is logged in and redirected to chatbot interface
Actual Result	Success
Status	Pass

Test Case 2: Chatbot Response

Test Case	Send a query to the chatbot
Expected Result	Relevant response generated by Gemini
Actual Result	Success
Status	Pass

Test Case 3: Chat History Saving

Test Case	Submit multiple messages during a session
Expected Result	All messages stored in MongoDB
Actual Result	Success

Test Case	Submit multiple messages during a session
Status	Pass

Test Case 4: API Failure Handling

Test Case	Disable internet and ask a question
Expected Result	Show error message gracefully
Actual Result	Success
Status	Pass

8.4 Tools Used for Testing

- Python's unittest framework
- Postman for API testing
- Manual testing via Streamlit interface
- MongoDB Compass and SQLite Browser for database verification

8.5 Bug Fixes and Improvements

- Fixed occasional session ID collision in SQLite
- Improved response formatting for large answers
- Added error handling for Gemini Pro API timeouts

8.6 Conclusion

- The system passed all major test cases and performs reliably under normal conditions. Through extensive testing, the chatbot was validated to be **user-friendly, responsive, and stable**, ensuring high-quality performance.

CHAPTER: 9

SNAPSHOT OF WEBSITE

9.1 How to run a server

9.2 Main Page of our site

9.3 Login/Sign Up page

9.4 Admin Panel

9.5 Chat Analysis

9.6 Example

9.7 Settings

9.1 How To Run a Server

- This is the main command of how to run a server of this project

```

app1.py > input, col1
1 import streamlit as st
2 import google.generativeai as genai
3 import json
4 import os
5 from datetime import datetime
6 import pytz
7 from database import init_database, get_course_data, save_chat, get_or_create_user_session
8
9 # ----- User Authentication -----
10 USERS_FILE = "users.json"
11
12 def load_users():
13     if not os.path.exists(USERS_FILE):
14         with open(USERS_FILE, "w") as f:
15             json.dump([], f)
16     with open(USERS_FILE, "r") as f:
17         return json.load(f)
18
19 def save_user(username, password):
20     users = load_users()
21     users[username] = password
22     with open(USERS_FILE, "w") as f:
23         json.dump(users, f)
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100

```

```

PS C:\Users\raha\Downloads\university-chatbot> venv\scripts\activate
(venv) PS C:\Users\raha\Downloads\university-chatbot> streamlit run app1.py

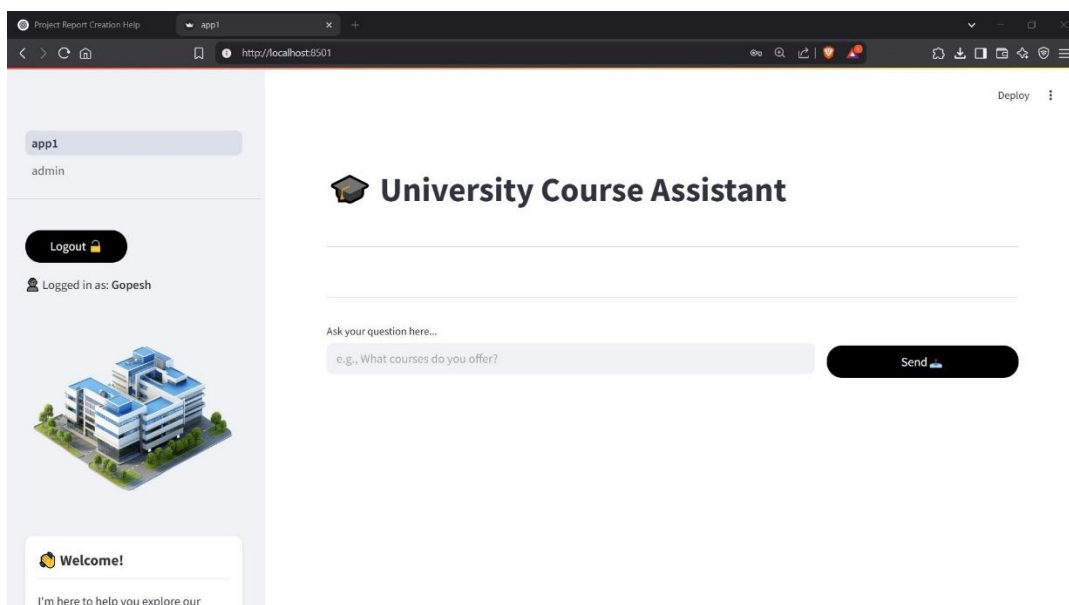
You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.10:8501

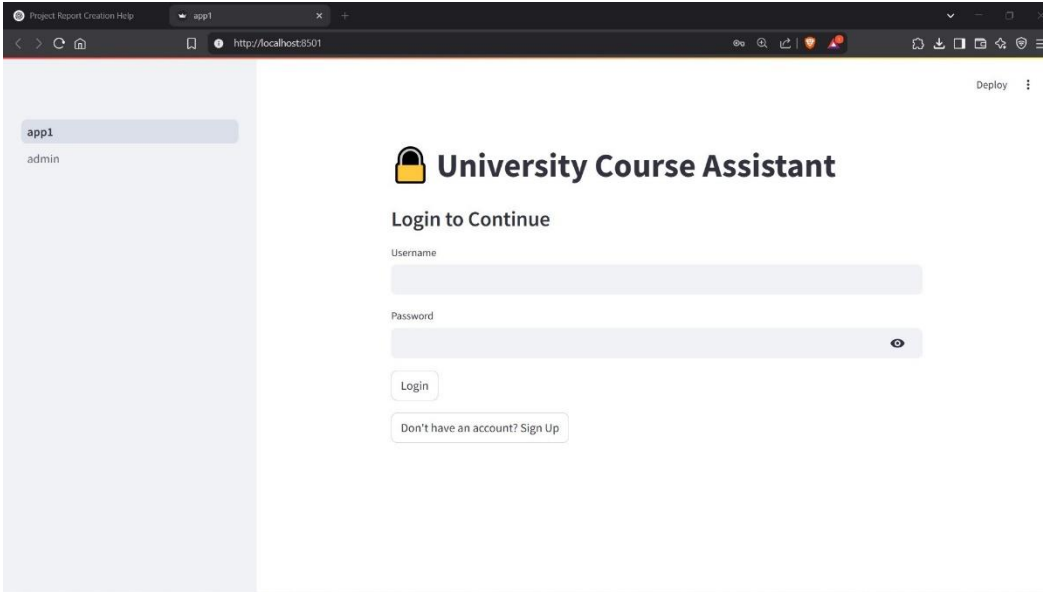
```

9.2 Main Page of Our Site

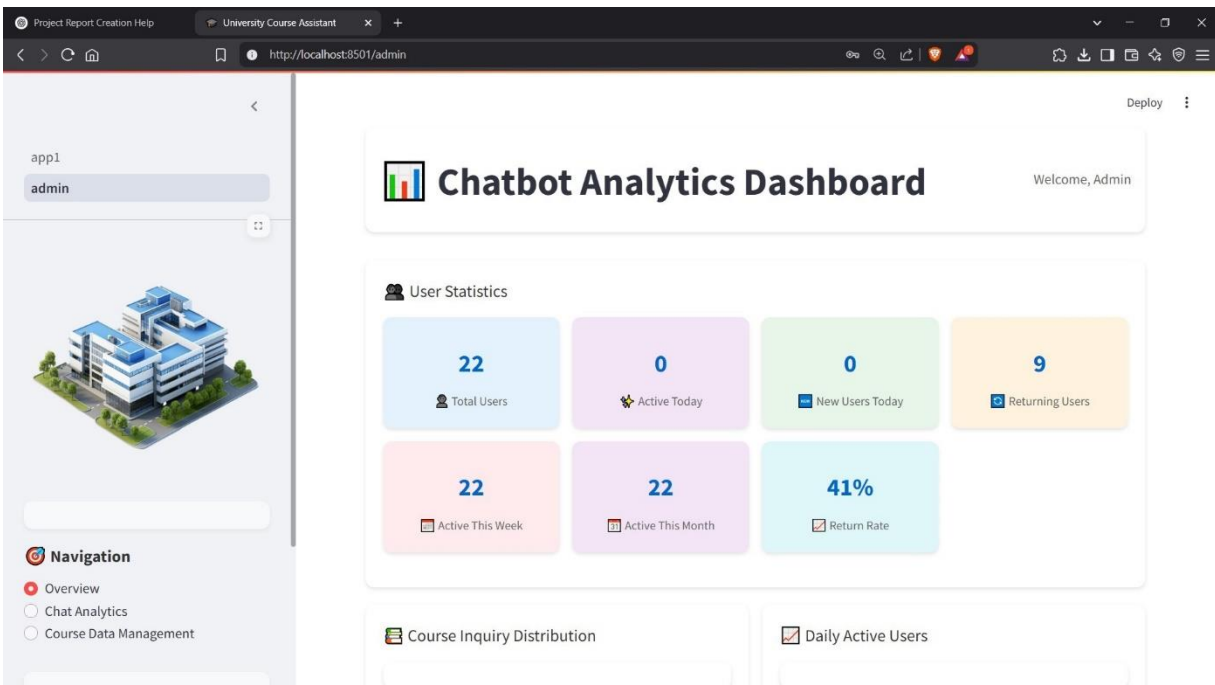
- This is the main page of our site new user can first create their account by clicking the SignUp page and the existing User can Login to their account using The Signin page.



9.3 Login/Sign Up Page



9.4 Admin panel



9.5 Chat Analysis

app1
admin

2025/03/06 2025/04/05

Recent Conversations

	timestamp	user_message	bot_response
0	2025-04-04 12:00:40	How long is the B.Tech program?	The B.Tech program is a 4-year course, consisting of 8 semesters.
1	2025-04-04 12:00:32	What is the fee structure for BCA?	The BCA program has a fee of ₹50,000 INR per semester. It's a 3-year p
2	2025-04-04 11:59:43	Hi! Can you help me with course information?	Hello! I'd be happy to help you with information about our B.Tech, B.
3	2025-04-04 11:37:02	Hi! Can you help me with course information?	Hello! I'd be happy to help you with information about our B.Tech, B.
4	2025-04-04 09:05:34	how are you?	I'm doing well, thank you for asking! I'm ready to assist you with any
5	2025-04-04 09:05:18	What courses do you offer?	Hello there! I'm happy to help you with information about the course
6	2025-04-04 08:26:40	What is the fee structure for BCA?	Hello! I'd be happy to help you with information about our courses. 1
7	2025-04-04 08:11:50	I'm Gopesh	Hello Gopesh! It's great to meet you! How can I help you with inform
8	2025-04-04 08:06:39	what is in B.Tech?	The B.Tech program is a 4-year course divided into 8 semesters. The fi
9	2025-04-04 08:06:03	What courses do you offer?	Hello! I can provide information about the following courses: * B.T

Download Chat History

9.6 Example

Project Report Creation Help app1

http://localhost:8501

Deploy

Example Questions

- Hi! Can you help me with course information?
- What courses do you offer?
- Tell me about B.Tech program
- What is the fee structure for BCA?
- What subjects are taught in B.Sc first semester?
- How long is the B.Tech program?
- What are the subjects in BCA?
- Tell me about admission process
- What is the duration of B.Sc?
- Can you compare B.Tech and BCA

University Course Assistant

You:
What subjects are taught in B.Sc first semester?

05:03

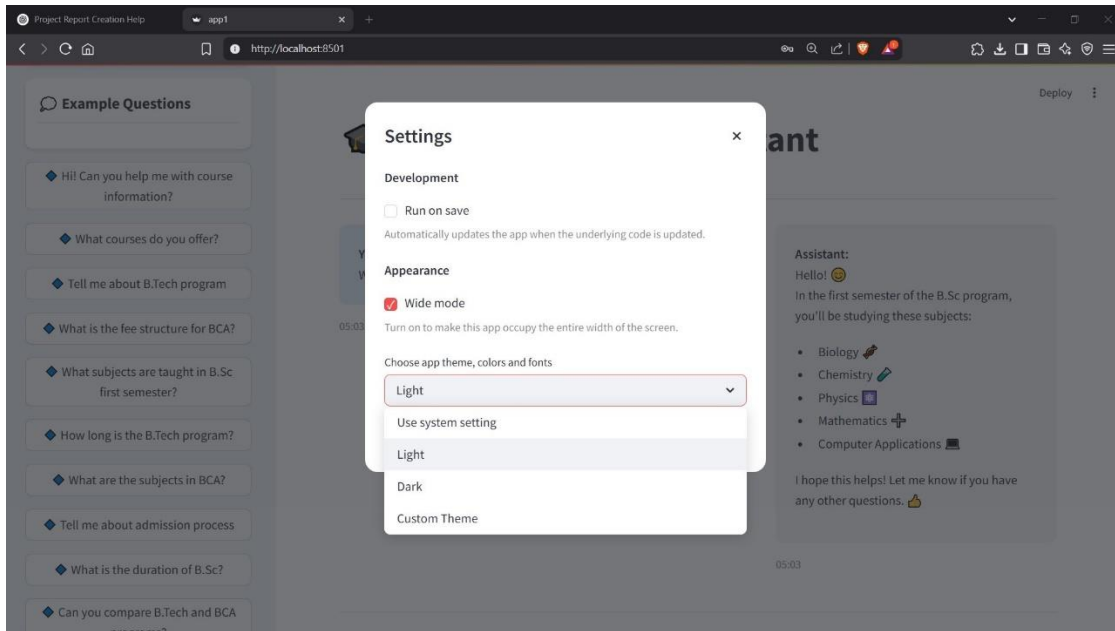
Assistant:
Hello! 😊
In the first semester of the B.Sc program, you'll be studying these subjects:

- Biology 🧬
- Chemistry 🧪
- Physics 📊
- Mathematics ➕
- Computer Applications 💻

I hope this helps! Let me know if you have any other questions. 🙌

05:03

9.7 Settings



CHAPTER: 10

ADVANTAGES

10.1 Advantages

10.2 Limitations

10.1 Advantages

1. 24/7 Availability

The chatbot can answer student queries at any time, even outside of regular office hours.

2. Reduced Human Workload

Handles frequently asked questions, reducing the burden on administrative staff and faculty.

3. Fast Response Time

Immediate answers are provided using the Gemini Pro API, enhancing user experience.

4. User-Friendly Interface

Built using Streamlit, the interface is simple, clean, and easy to navigate for students.

5. Session-Based Chat Support

Maintains chat history per session, allowing more natural conversations.

6. Secure Data Storage

Uses MongoDB for scalable cloud storage and SQLite for local session data, with secure login support.

7. Smart and Contextual Responses

Integration with Google Gemini Pro ensures intelligent and relevant responses to a wide variety of academic queries.

10.2 Limitations

1. Limited to Pre-Trained Knowledge

- The chatbot relies on Gemini's training and may not answer very specific or university-unique queries unless customized further.

2. Requires Internet Access

- Since Gemini API and MongoDB Atlas are cloud-based, the system does not work offline.

3. No Voice Input/Output

- Currently supports only text-based interaction. Voice support is not implemented.

4. No Real-Time Admin Monitoring

- Admins cannot monitor ongoing chat sessions live unless additional features are built.

5. Dependency on Third-Party APIs

- If the Gemini API service is down or restricted, the chatbot loses its primary function.

6. Basic Security Implementation

- While login/signup is in place, more robust features like two-factor authentication or role-based access can enhance security.

CHAPTER: 11

CONCLUSION AND FUTURE ENHANCEMENT

11.1 Conclusion

11.2 Future Enhancement

11.1 Conclusion

- The **University Chatbot** project successfully demonstrates how artificial intelligence can enhance student support and academic communication. By integrating modern technologies like **Google Gemini Pro**, **Streamlit**, **MongoDB**, and **SQLite**, the system offers a responsive, intelligent, and user-friendly experience.
- It effectively simulates a human-like assistant capable of answering university-related questions, reducing staff workload and increasing accessibility. The chatbot ensures consistent and quick responses, offering students a reliable medium to resolve their academic queries at any time.

The project fulfilled its main objectives:

- Developed an AI-based chatbot system.
- Enabled user login and session-based chat.
- Stored chat histories securely in cloud and local databases.
- Delivered meaningful and relevant responses via Gemini API.

11.2 Future Enhancements

While the system achieves its core functionality, there are several enhancements that can be explored in the future:

1. Voice Input and Output

- Adding voice-based interaction to improve accessibility and ease of use.

2. Admin Dashboard

- A real-time monitoring panel for tracking usage analytics, chatbot performance, and user queries.

3. Integration with University Database

- Connect with official databases to answer personalized academic queries like results, fees, class schedules, etc.

4. Role-Based Access

- Implement different roles like student, admin, and faculty for customized features.

5. Multilingual Support

- Enable interaction in regional or foreign languages for a broader user base.

6. Offline Mode

- Adding limited offline functionality for use in areas with poor internet connectivity.

7. Auto-Learning Mechanism

- Let the chatbot learn from past chats to improve future responses (requires advanced NLP fine-tuning).

BIBLIOGRAPHY

a. Course Outcome

b. Books

c. Web Reference

a. Course Outcome

Sr. No.	CO statement	Marks % weightage	Chapter No.
CO-1	Undertake problem identification, formulation and solution	20%	CHAPTER 1 INTRODUCTION
CO-2	Design engineering solutions to complex problems utilising a systematic approach and team work	30%	CHAPTER 2 SYSTEM REQUIREMENTS CHAPTER 3 DAILY TASK
CO-3	Communicate with engineers and the community at large in written and oral forms	20%	CHAPTER 6 SYSTEM DESIGN CHAPTER 10 ADVANTAGES CHAPTER 11 CONCLUSION AND FUTURE ENHANCEMENT
CO-4	Demonstrate the knowledge and understanding of engineering and management principle and apply it to assigned project	30%	CHAPTER 4 FRONT END OF SYSTEM CHAPTER 5 BACK END OF SYSTEM CHAPTER 6 SYSTEM DESIGN

			CHAPTER 7 DATA DICTIONARY CHAPTER 8 TESTING
--	--	--	--

b. Books:

1. **"Artificial Intelligence: A Guide to Intelligent Systems"** by Michael Negnevitsky
 - This book provides a comprehensive introduction to AI and intelligent systems, including natural language processing (NLP) and chatbot development. It is a valuable resource for understanding the theoretical foundations of AI and its practical applications.
2. **"Natural Language Processing with Python"** by Steven Bird, Ewan Klein, and Edward Loper
 - This book is a practical guide to NLP using Python, with a focus on the Natural Language Toolkit (NLTK). It covers essential NLP techniques that are relevant to chatbot development, such as tokenization, parsing, and sentiment analysis.
3. **"Building Chatbots with Python"** by Sumit Raj
 - This book is a hands-on guide to building chatbots using Python. It covers the basics of chatbot development, including NLP, machine learning, and integration with messaging platforms. It is particularly useful for beginners in chatbot development.
4. **"Deep Learning for Natural Language Processing"** by Palash Goyal, Sumit Pandey, and Karan Jain

- This book explores advanced NLP techniques using deep learning. It covers topics such as word embeddings, sequence-to-sequence models, and transformer architectures, which are essential for building sophisticated chatbots.

5. **"Python Machine Learning"** by Sebastian Raschka and Vahid Mirjalili

- This book provides a comprehensive introduction to machine learning with Python. It covers key machine learning algorithms and techniques that can be applied to chatbot development, such as classification, clustering, and neural networks.

6. **"Designing Bots: Creating Conversational Experiences"** by Amir Shevat

- This book focuses on the design and user experience aspects of chatbot development. It provides insights into creating engaging and effective conversational interfaces, which are crucial for the success of any chatbot project.

c. Web Reference:

➤ **W3Schools Python Tutorial**

<https://www.w3schools.com/python/>

A comprehensive tutorial for learning Python, covering basic to advanced topics. It is a useful resource for understanding the programming language used in the chatbot project.

➤ **Towards Data Science (Medium)**

<https://towardsdatascience.com/>

A platform for articles and tutorials on data science, machine learning, and AI. It includes many articles on chatbot development and NLP.

➤ **Chatbots Magazine (Medium)**

<https://chatbotsmagazine.com/>

A publication focused on chatbot development, featuring articles, tutorials, and case studies. It is a valuable resource for staying updated on the latest trends and best practices in chatbot development.

➤ **Google AI Blog**

<https://ai.googleblog.com/>

The official blog for Google AI, featuring articles on the latest advancements in AI and machine learning. It includes insights into NLP and chatbot technologies.

➤ **OpenAI Documentation**

<https://platform.openai.com/docs/>

The official documentation for OpenAI, including the GPT models. It provides resources for integrating advanced language models into chatbot applications.

➤ **Kaggle**

<https://www.kaggle.com/>

A platform for data science and machine learning competitions, datasets, and tutorials. It includes many resources for NLP and chatbot development.

➤ **Coursera - Natural Language Processing Specialization**

<https://www.coursera.org/specializations/natural-language-processing>

A series of online courses on NLP offered by leading universities. It covers key concepts and techniques relevant to chatbot development.

➤ **edX - Artificial Intelligence (AI) Courses**

<https://www.edx.org/learn/artificial-intelligence>

A collection of online courses on AI and machine learning, including NLP and chatbot development. It is a valuable resource for learning advanced AI techniques.

- **Other**

- Streamlit Documentation – <https://docs.streamlit.io/>
- Google Gemini Pro API – <https://deepmind.google/technologies/gemini/>
- Python Official Documentation – <https://docs.python.org/3/>
- MongoDB Documentation – <https://www.mongodb.com/docs/>
- SQLite Documentation – <https://www.sqlite.org/docs.html>
- Stack Overflow – <https://stackoverflow.com/>
- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- W3Schools – <https://www.w3schools.com/>
- Real Python – <https://realpython.com/>